

Robot Programming in TinkerCAD- Human Gripper



Valai Ganesh • Aug 27 (Edited Aug 28)

100 points

Due Sep 3, 11:59 PM

Dear Students,

Herewith I have attached the Presentation file for your reference. Go through it.

The question is as follows:

Add ultrasonic sensor in front of the gripper. If any object/signal received means all fingers will contract and extend happens in parallel actions. Develop a program in Tinkercad using Arduino, Ultrasonic sensor, Servo motors etc.

If you need any clarifications, please let me know.

Instructions:

- (a) Maximum 3 members per team
- (b) Team Formation should be mentioned prior via Whats App group
- (c) Team who completed and submitted first with all those conditions will be honored
- (d) Last date for submission is 03 September 2020 at 11:59 pm. We will discuss the answers in 05 September 2020 Class

 Rubric: 4 criteria • 100 pts



Robotic Grasping Arm (2).pdf
PDF

Class comments



Add class comment...



 Rubric**GROUP CHALLENGE- RUBRICS**

Only text edits are allowed once you've started grading with a rubric

Robot Programming in TinkerCAD- Human Gripper

/100



Usage of Components

/25



Place the components in correct order

I	25 pts	II	20 pts	III	14 pts
Excellent		Good		Average	

Programming with conditions

/25



Create the program with all conditions

I	25 pts	II	20 pts	III	15 pts
Excellent		Good		Average	

Aesthetic look

/25



Place the components and connections in a proper look

III	25 pts	II	20 pts	I	15 pts
Excellent		Good		Average	

Submission

/25



Submission

I	25 pts	II	20 pts	III	15 pts
Early Submission with all conditions		Early Submission with 90% conditions		Early Submission with 80% conditions	



ROBOTIC GRASPING ARM

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ABSTRACT

Design the gripper like a human hand for handling the objects in pick and place robot applications. And also attain motions like a human hand during the grasping of objects. And obtain various pressure controls in order to grasp the complex objects for different nature of objects.

NEED FOR

IMPROVEMENTS



ADVANCEMENT

Need of advancement in gripping operation

CONTROL

Multifunctional gripping under one product with different mode of controlling system

SIMPLIFICATION

Reducing components and complexities without compromising desired operation

CAD MODELS

IMPORTANT PARTS

Assembled model



Distal



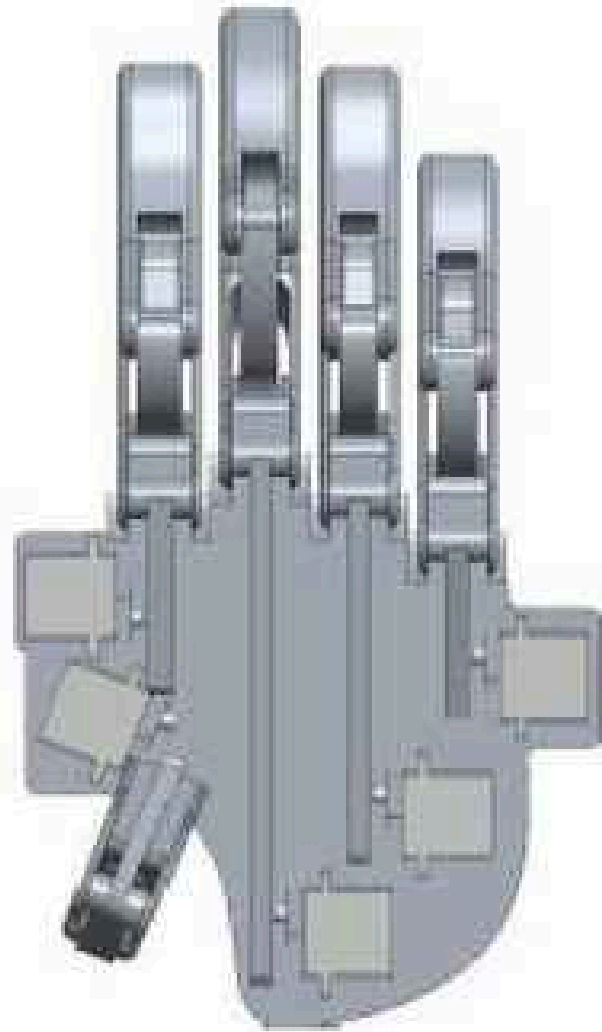
Interlink



Proximal



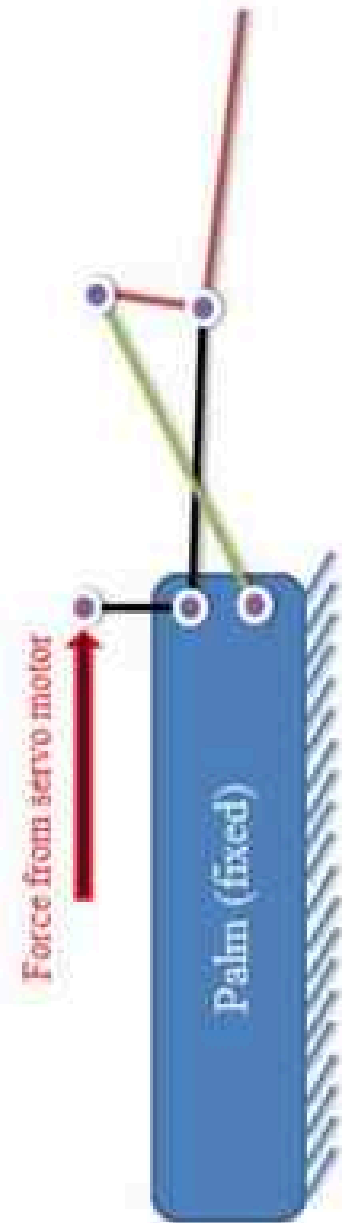
ROBOTIC GRASPING ARM



POSITIONS OF MOTORS

ROBOTIC GRASPING ARM

BACKSIDE VIEW | DESIGNED IN CREO PARAMETRIC



SCHEMATIC MODEL OF A FINGER

Kutzbach criterion:

$$\begin{aligned}\text{DOF} &= 3(L-1) - 2J - H \\ &= 3(3) - 2(4) - 0 \\ &= 1\end{aligned}$$

Degree of freedom = 1

Total degrees of freedom = 5

COMPONENTS

USED

Servo motor

1.8 Kgf cm.
4.8-5V



Arduino UNO

5V
14(Digital),
6 (Analog) I/O pins



Force sensor

0-1.8kg
Sensing Area-3.8 mm



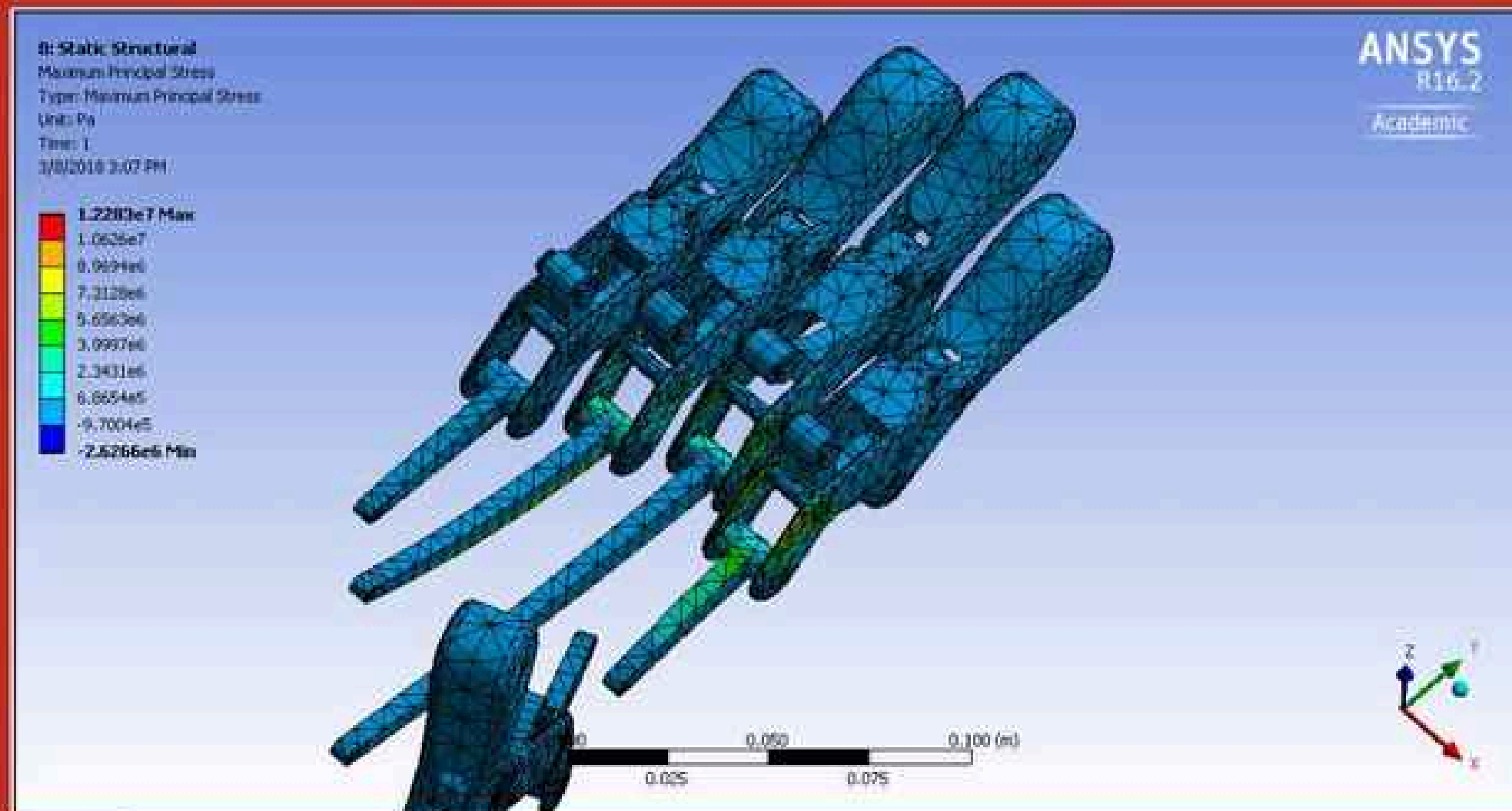
MANUFACTURING OF GRIPPER 3DPRINTING



Material : PLA
Method : FDM
Max Strength : 37MPa

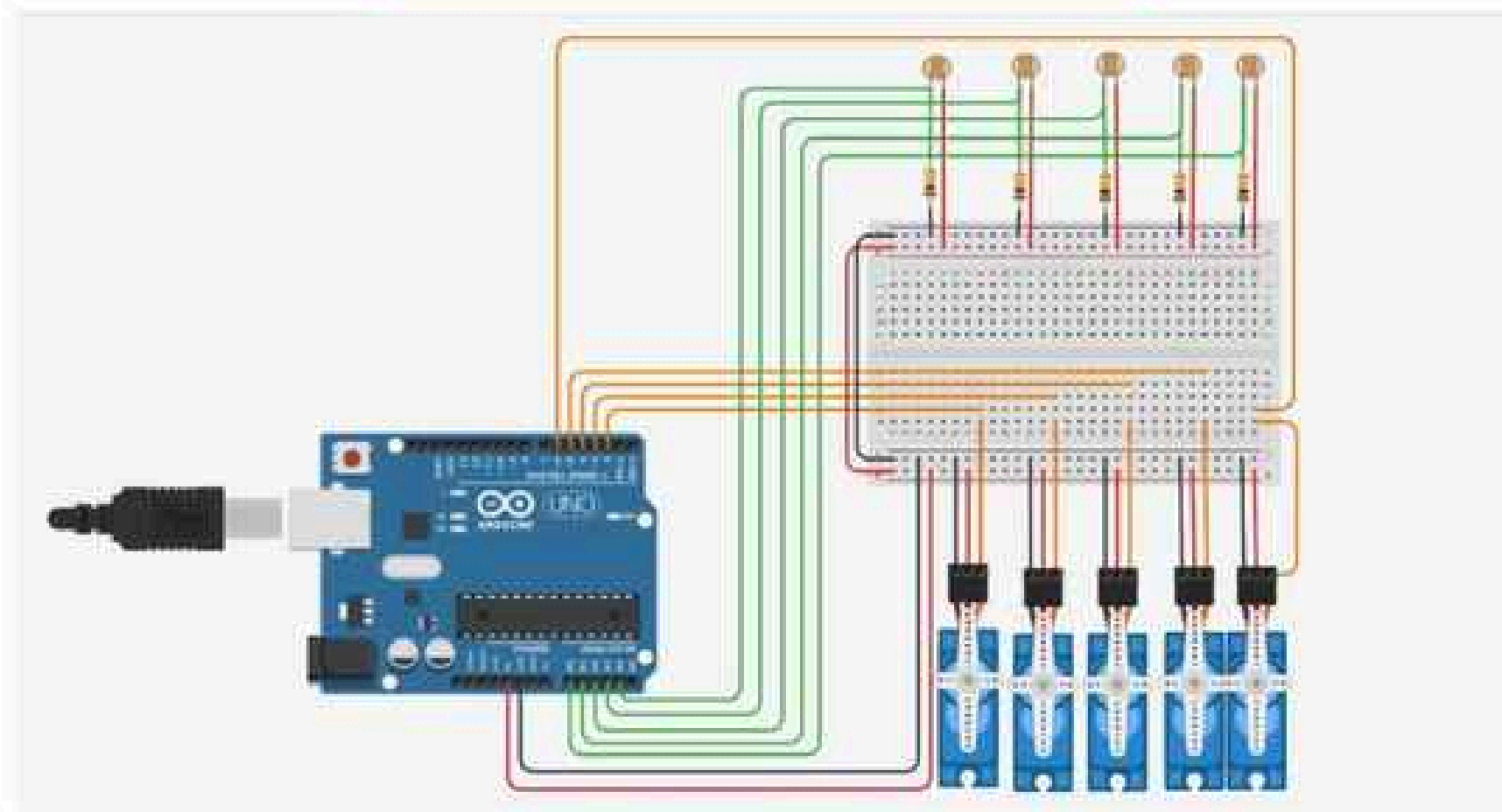
ANALYSIS

MAXIMUM PRINCIPLE STRESS

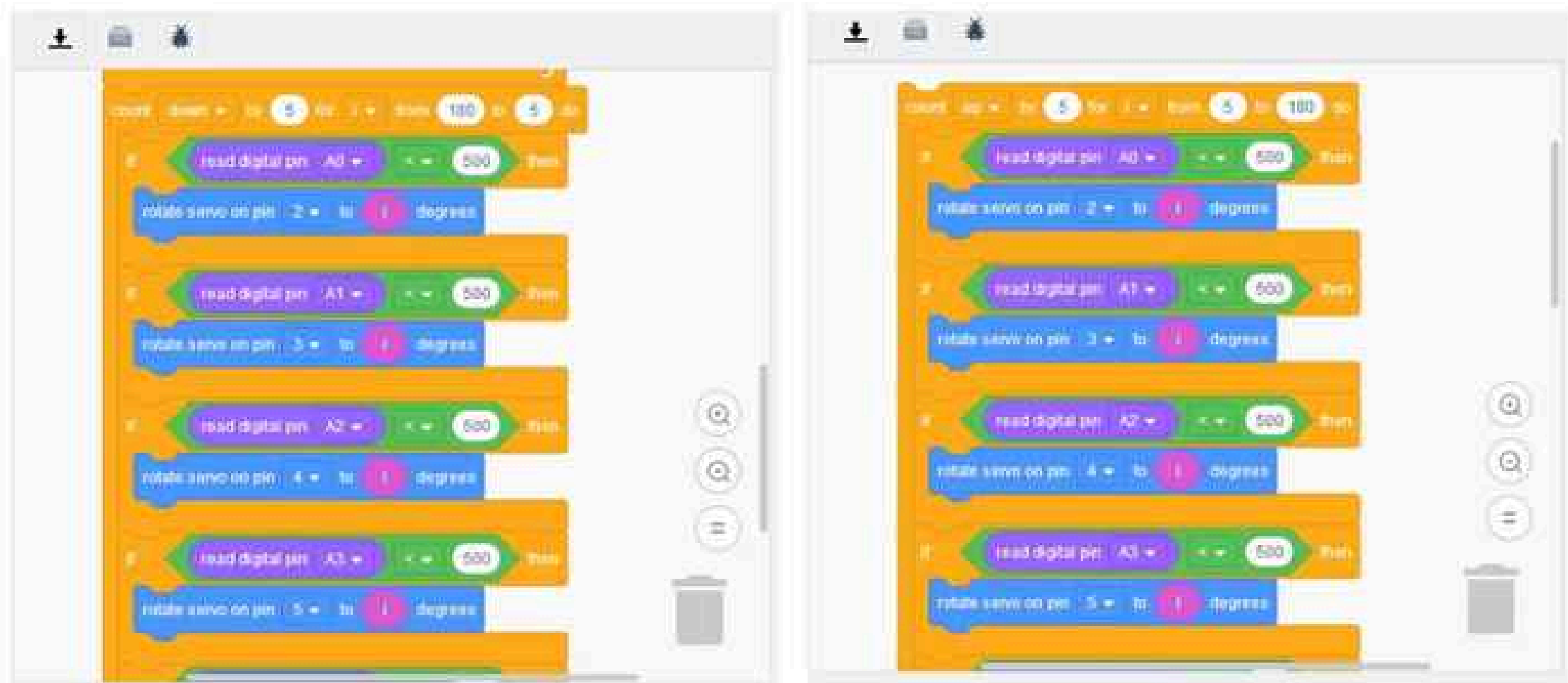


CIRCUIT

CONNECTION



AUTODESK TINKERCAD



ARDUINO PROGRAMMING

```
void setup()
{
  pinMode(A0, INPUT);
  servo_2.attach(2);

  pinMode(A1, INPUT);
  servo_3.attach(3);

  pinMode(A2, INPUT);
  servo_4.attach(4);

  pinMode(A3, INPUT);
  servo_5.attach(5);

  pinMode(A4, INPUT);
  servo_6.attach(6);
}
```

```
void loop()
{
  for (i = 5; i <= 180; i += 5) {
    if (digitalRead(A0) < 500) {
      servo_2.write(i);
    }
    if (digitalRead(A1) < 500) {
      servo_3.write(i);
    }
    if (digitalRead(A2) < 500) {
      servo_4.write(i);
    }
    if (digitalRead(A3) < 500) {
      servo_5.write(i);
    }
    if (digitalRead(A4) < 500) {
      servo_6.write(i);
    }
    delay(3000); // Wait for 3000 milliseconds
  }
  for (i = 180; i >= 5; i -= 5) {
    if (digitalRead(A0) < 500) {
      servo_2.write(i);
    }
    if (digitalRead(A1) < 500) {
      servo_3.write(i);
    }
    if (digitalRead(A2) < 500) {
      servo_4.write(i);
    }
  }
}
```

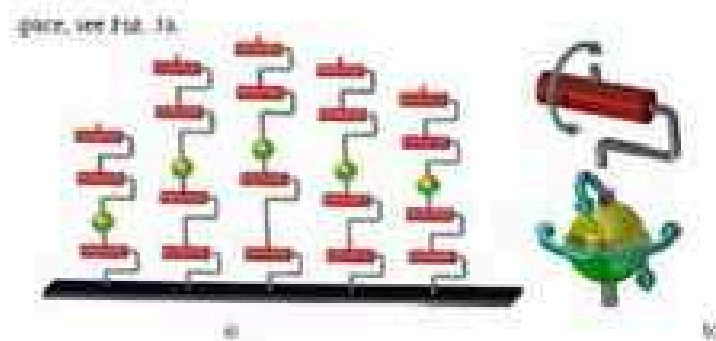


Fig. 1. a) Kinematic configuration of human hand, b) available joint motion.



Fig. 2. Project overview of controlling cyber arm.



LITERATURE

SURVEY

Analyzing, Modeling and Simulation of Humanoid Robot Hand Motion(2012) -Ivan Virgala, Michal Kelemen, Martin

Wireless Mobile Robotic Arm
(2012) -Mohd Ashiq Kamaril Yusoff, Ezuan Samin,

Dynamic Modeling and Control of a Multi-Fingered Robot Hand for Grasping Task (2014) -
RimBoughdiria , Habib Nasser , Hala Bezine

CONCLUSION

THE ROBOTIC GRASPING ARM CAN BE USED AS A END EFFECTOR OF A ROBOTS. ITS WORKING CAN BE MODIFIED WITH THE CONTROLLING SYSTEM AND PROGRAMMING FOR REQUIREMENTS.

THANK YOU!



